

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (ME) Examination

DEC 2017

ME417 Internal Combustion Engines

Date: 01.12.2017, Friday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Answer the question below.

[05]

- a.** The ratio of indicated thermal efficiency to the corresponding air standard cycle efficiency is called
1. Net efficiency
 2. Efficiency ratio
 3. Relative efficiency
 4. Overall efficiency
- b.** For same compression ratio and heat supplied, the air standard efficiency of an otto cycle compared to that of a diesel cycle is
1. Less
 2. More
 3. Equal
 4. Unpredictable
- c.** The petrol engine are usually not supercharged as that results in
1. Increased knocking
 2. Power loss
 3. Increased specific fuel consumption
 4. Drop in volumetric efficiency
- d.** HUCR is the compression ratio at which the engine
1. Gives maximum power output
 2. Consumes minimum fuel for a particular power output
 3. Maintains operating pressures and temperatures within prescribed limits
 4. Can operate without detonation
- e.** Which of the following is not a bad effect of NO_x ?
1. Damage of crops
 2. Affects animal life
 3. Highly hazardous to public health
 4. Reduces visibility.

Q-2.(a) Explain effects of volatility on SI engine performance.

[05]

- Q-2.(b)** A Petrol engine of compression ratio 6 uses a fuel of calorific value 44000 kJ/kg. The air-fuel ratio is 15:1. The temperature and pressure of the charge at the end of the suction stroke are 60°C and 1 bar, respectively. Determine the maximum pressure in the cylinder if the index of compression is 1.32 and the specific heat at constant volume is expressed by the relation, $c_v = 0.71 + 20 \times 10^{-5} T$ kJ/kgK where T is the temperature in K. compare this value with that of constant specific heat $c_v = 0.71$ kJ/kgK. **[06]**

OR

- (b)** Explain variation in specific heat and losses incurred due to it in otto cycle. **[06]**
- (c)** Discuss factors affecting flame propagation in SI engines **[06]**

OR

- (c)** What are requirements of CI combustion chamber for diesel engines? Explain why weak mixtures give better efficiency in CI engines. **[06]**

- Q-3.(a)** Explain phenomenon of pre ignition in SI engine. Does it cause knocking in the engines? Justify. **[07]**

OR

- (a)** What are the causes of knock in CI engine? During which part of combustion process does it occur? Give its effects on engine performance. **[07]**
- (b)** Explain eddy current dynamometer along with neat sketch, merits and demerits. **[06]**

OR

- (b)** Explain effects of different pollutants on humane and plant life. **[06]**

SECTION – II

- Q - 4** Answer the question below. **[05]**

- a.** In CRDI engines, HP pump supplies fuel to
- | | |
|------------------|-----------------------------------|
| 1. LP pump | 2. Nozzle |
| 3. Common header | 4. Collapsed Rim Diesel Injection |
- b.** The minimum number of rings in a piston are
- | | | | |
|------|------|------|------|
| 1. 1 | 2. 2 | 3. 3 | 4. 4 |
|------|------|------|------|

c. In 2 stroke 2 cylinder CI engine, running at 3000 rpm number of sparks occurring in 5 minutes are

1. 150000 2. 7500 3. 30000 4. none

d. Horsepower is a measure of

1. The amount of work done. 2. The speed of shaft rotation.
3. The rate of doing work. 4. The amount of heat generated.

e. Which of the following nozzles have a better cold starting performance

1. Single hole 2. Multi hole
3. Pintle 4. Pintaux

Q-5.(a) Explain carter carburetor with neat sketch. **[07]**

OR

(a) Explain any three systems from main metering system of a carburetor **[07]**

- (i) Compensating jet device (ii) Emulsion tube or Air bleeding device
(iii) Back suction control (iv) Auxiliary valve

(b) Explain battery ignition system with neat sketch. **[07]**

OR

(b) Compare various types of injection system. Also discuss jerk type pump system. **[07]**

(c) Discuss the requirements of an Ideal injection. Enlist various injection nozzles. **[06]**

Q-6.(a) Explain turbocharging give advantages and disadvantages of constant pressure turbo charger. **[06]**

OR

(a) What is super charging how it is achieved? Explain the objects of supercharging. **[06]**

(b) Explain Wankel's engine **[04]**
